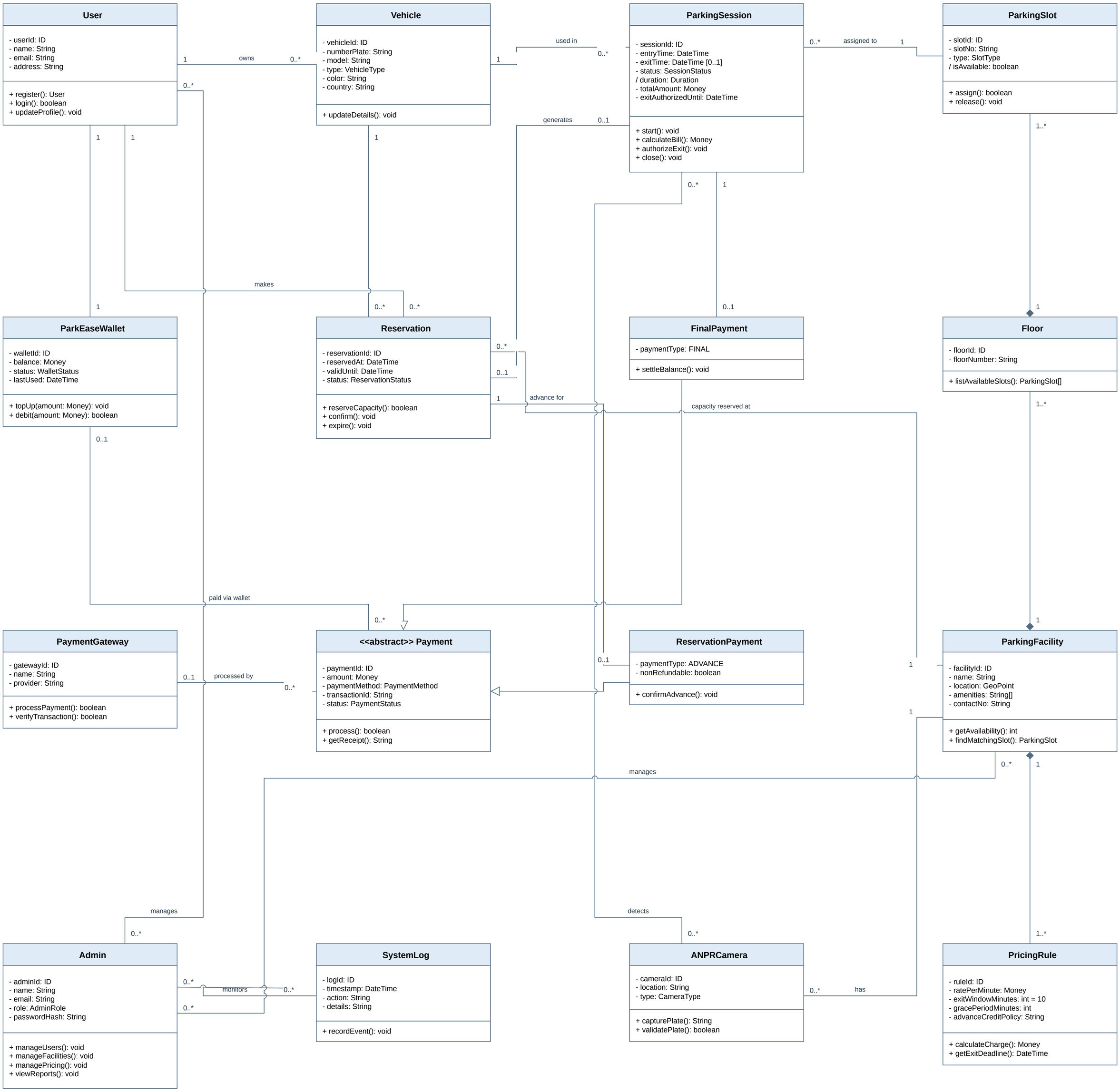
ParkEase - Complete UML Class Diagram

All 16 domain classes in one view | Attributes, operations, multiplicities and inheritance



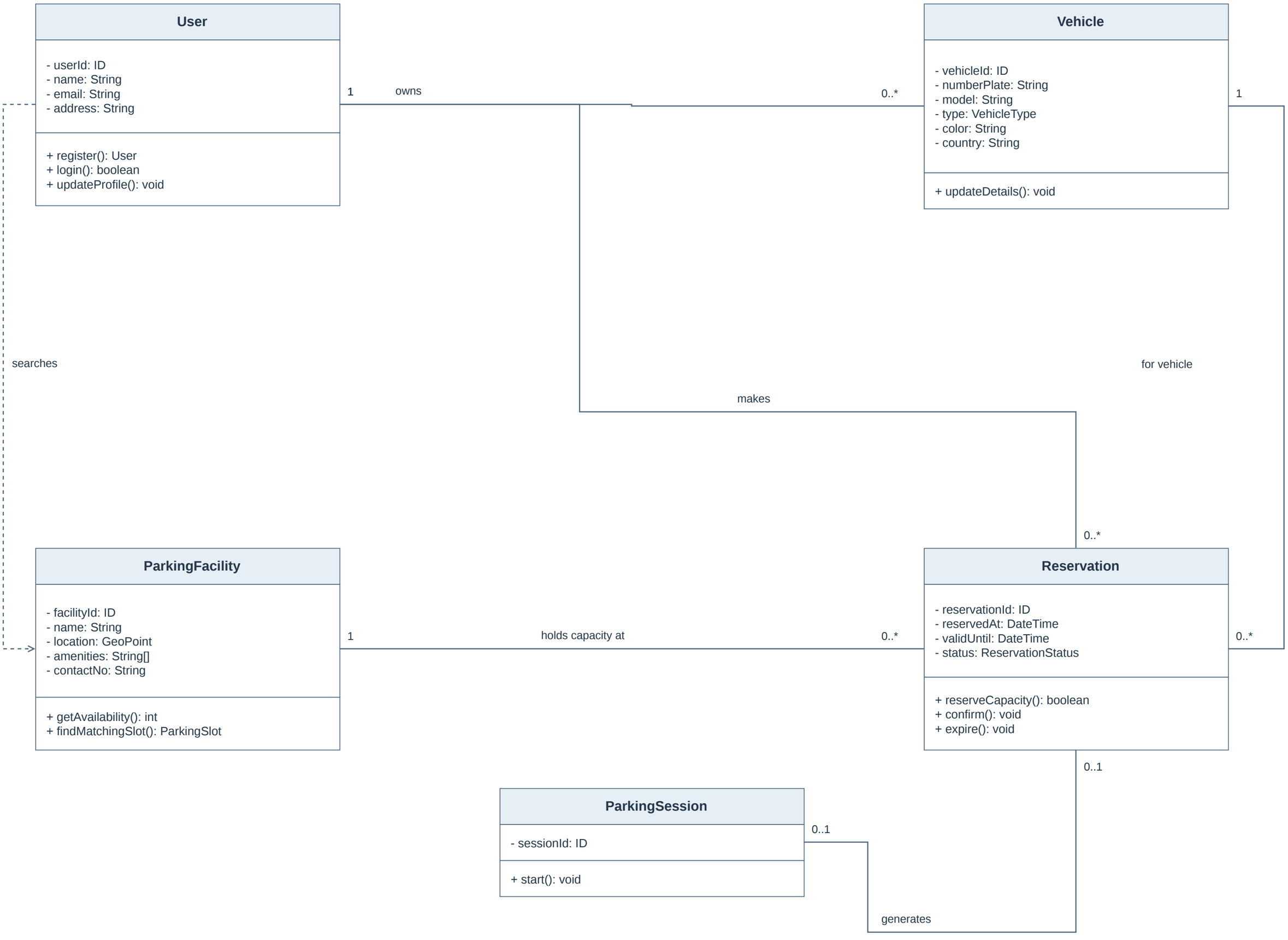
Rules: reserve capacity first; assign a physical slot after ANPR entry verification. One active session per occupied slot.

Payment uses either Wallet or Gateway. Final charge and advance credits follow PricingRule. Exit window: 10 minutes plus grace.

UML: + public method | - private attribute | / derived attribute | hollow triangle = inheritance | filled diamond = composition | 0..* = many

01 - Users and reservations

Core booking classes. Association lines show references; identifiers use the logical type ID.

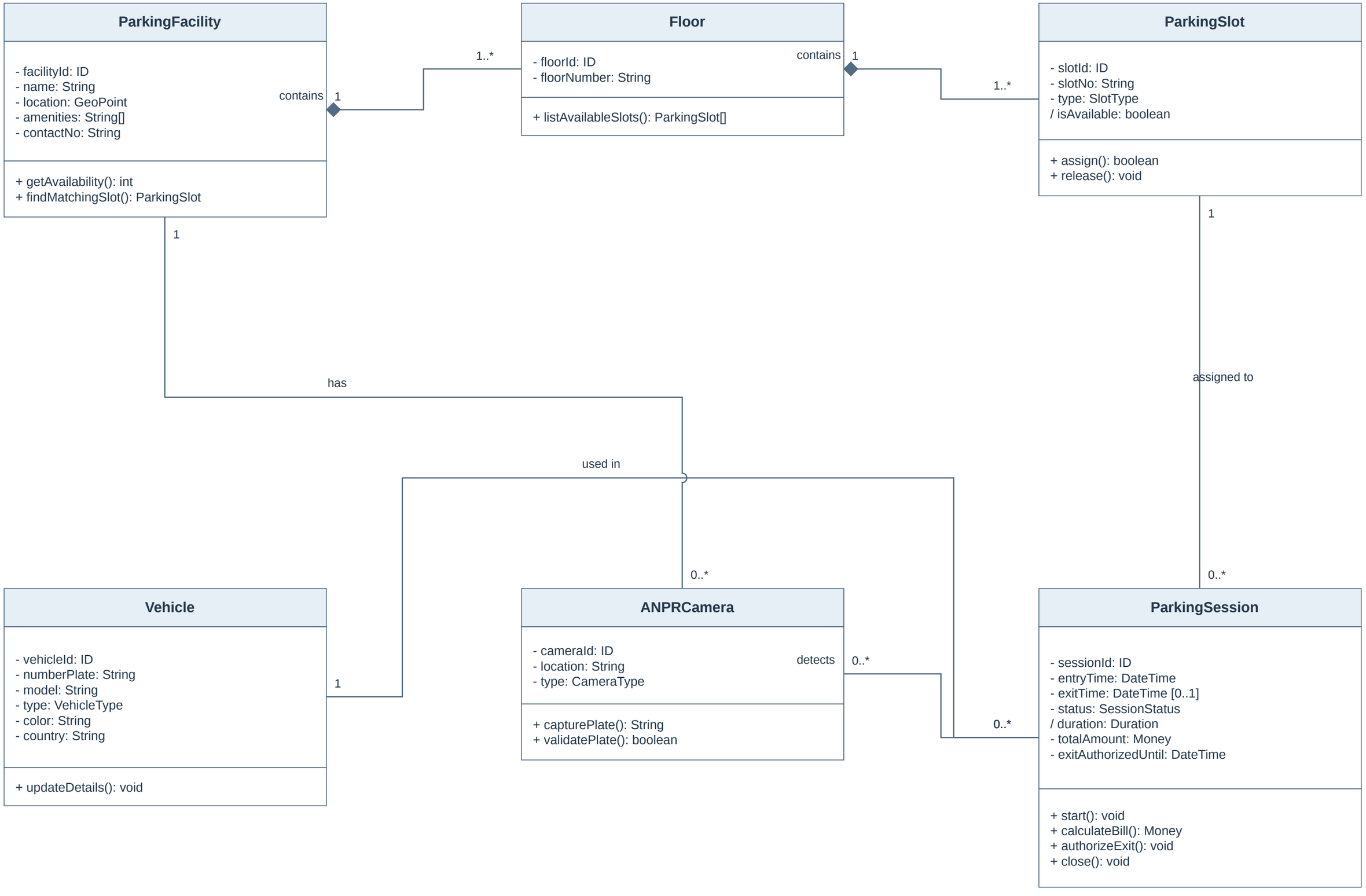


Reservation holds facility capacity; no physical slot is assigned until arrival. Classes repeated on other pages refer to the same class.

+ public method - private attribute / derived attribute 0..1 optional 0..* many hollow triangle: inheritance filled diamond: composition

02 - Parking and ANPR

Physical parking structure and session lifecycle. A slot can serve many sessions over time, with at most one active session.

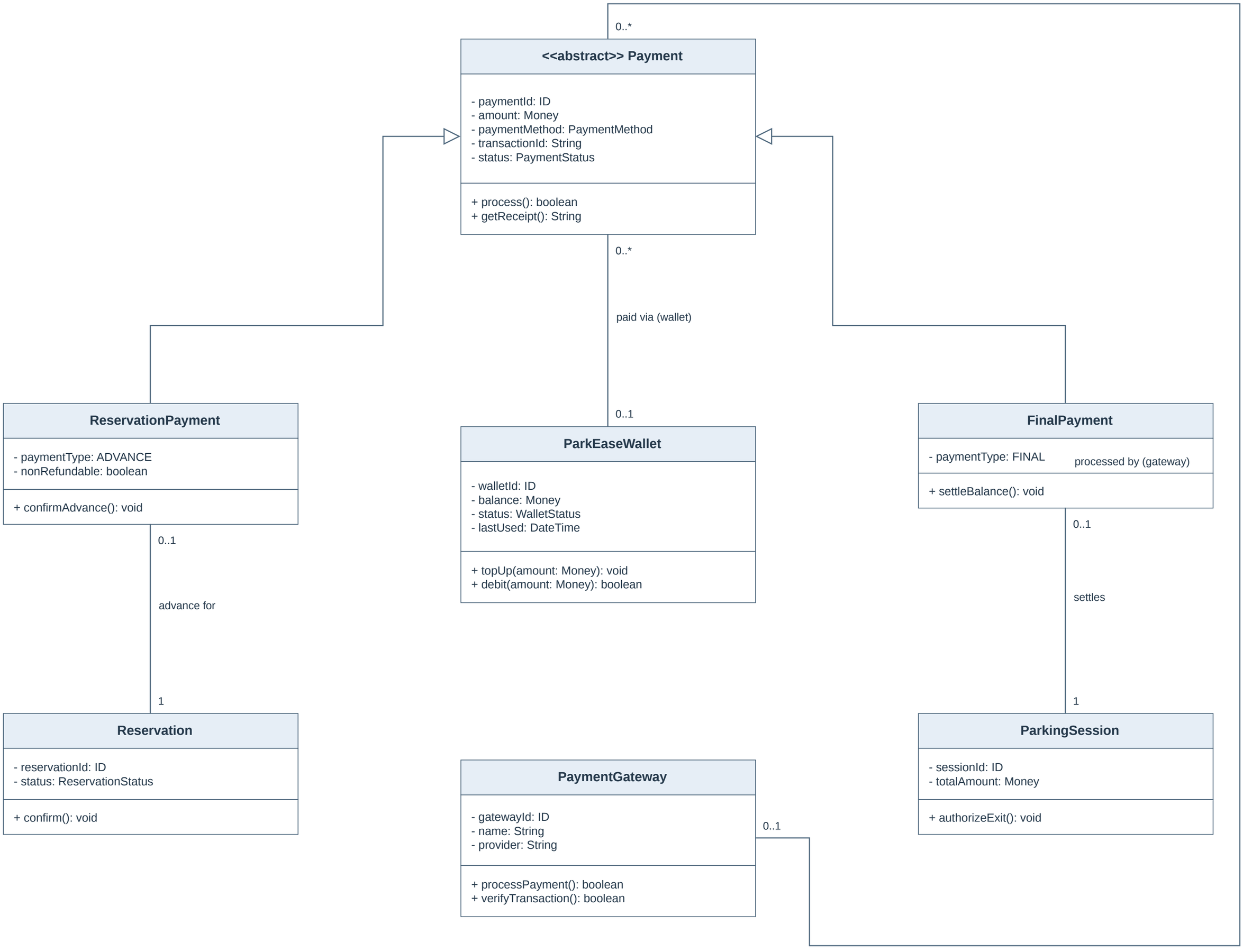


A session starts after vehicle verification and dynamic slot allocation. Successful payment grants a 10-minute exit window plus configurable grace.

+ public method - private attribute / derived attribute 0..1 optional 0..* many hollow triangle: inheritance filled diamond: composition

03 - Payments and wallet

Payment is a shared abstraction for ReservationPayment and FinalPayment. Amounts and settlement follow facility rules.

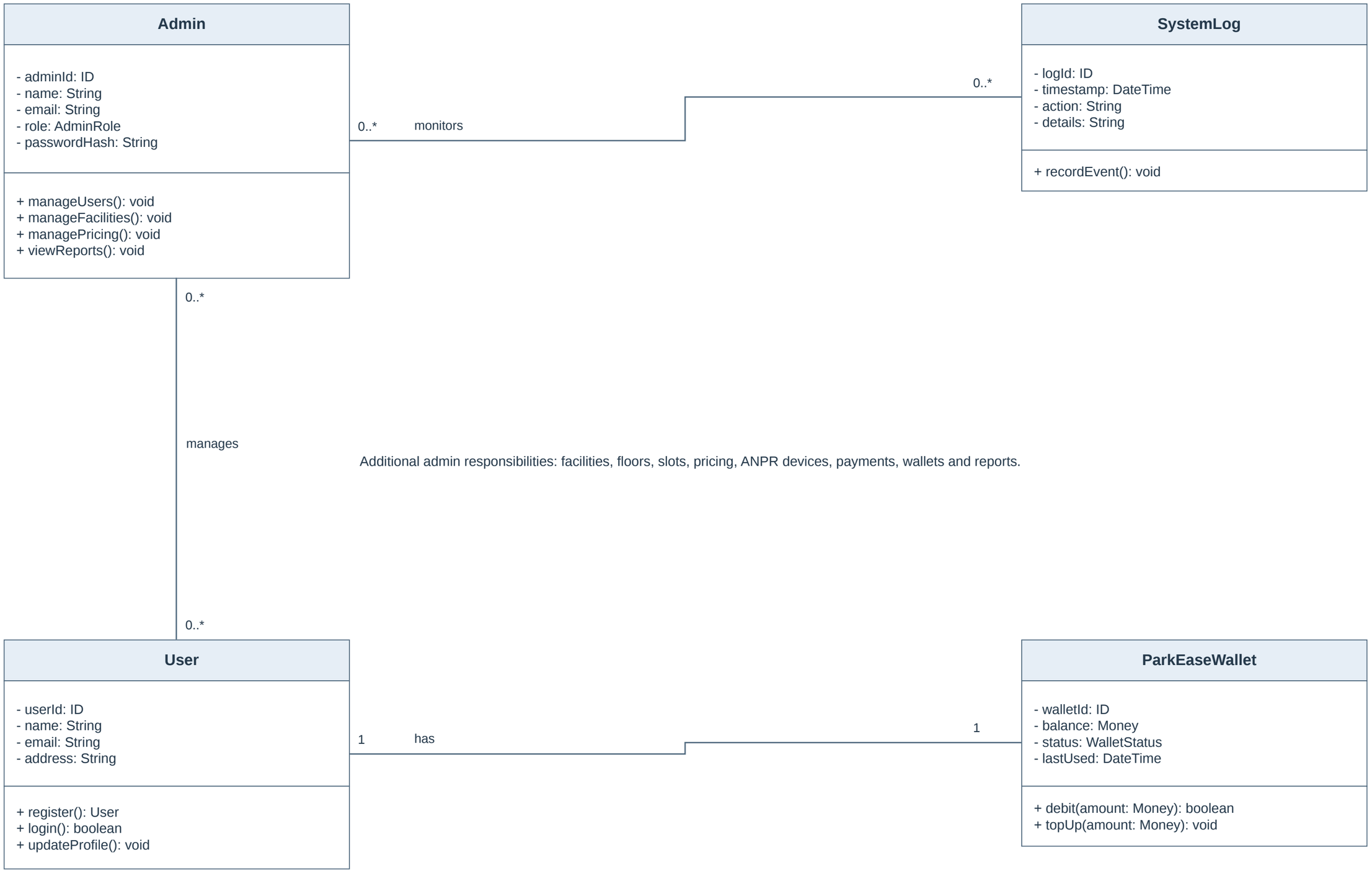


Each payment uses either a wallet or a gateway. A pending reservation/session can have no successful payment yet; failed attempts may be logged separately.

+ public method - private attribute / derived attribute 0..1 optional 0..* many hollow triangle: inheritance filled diamond: composition

04 - Administration and ownership

Administrative responsibilities and supporting records. Manage relationships denote responsibilities, not ownership of business records.



Design detail: passwords are represented as hashes. Floor, operation signatures and payment inheritance are class-design additions based on the project workflow.

+ public method - private attribute / derived attribute 0..1 optional 0..* many hollow triangle: inheritance filled diamond: composition